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Lead Backend & Data Science professional with 7+ years of experience in **AI, SaaS, and Fintech platform using Python**. **Expert in Natural Language Processing, Django/FastAPI, async and event-driven systems, cloud-native deployments, and GenAI/LLM-backed features.** Strong ownership of system design, release engineering, observability, and scaling zero-to-one products used by 10,000+ users.

WORK EXPERIENCE

- **Lead Backend and AI Platform Engineer**
  Enalo Technologies Pvt. Ltd., Hyderabad — Jun. 2022 – Dec 2025
  - Built an LLM-powered user story generation platform using **FastAPI and React**, reducing BA effort by **~65%**.
  - Designed RAG pipeline and prompt orchestration using LangChain, Ollama, and LangGraph, for deterministic, and high quality output.
  - Used MLFlow for model/prompt versioning/engineering, experiment tracking, and evaluation.
  - Developed quality scoring and validation pipeline to ensure stable, production-ready outputs
    - **Semantic cohesion scoring** to validate contextual integrity and logical completeness checks to detect logical drift and hallucinations.
    - **EARS-based scoring** for requirements clarity, actionability, specificity, and acceptance-criteria alignment
    - **LLM-as-Judge evaluation** for automated qualitative scoring and regressing testing.
  - Designed and implemented **RAG GenAI workflows** with guardrails for malformed outputs, prompt failures, and non-deterministic responses.
  - Implemented **stateless and vendor-agnostic LLM services** with pluggable model backends.
  - Tech Stack: Python, FastAPI, LangChain, LangGraph, Ollama, MLFlow, Huggingface, PineCone, Prompt Engineering, Natural Language Processing, Large Language Model, Message Queue, Event Driven Architecture
  - Architected and scaled a fintech SaaS platform for invoice automation, vendor payments, and reconciliations, supporting 10,000+ merchants.
  - Integrated **banking APIs (ICICI Bank, Yes Bank)**, payment gateway, and KYC services to automate end to-end payment workflows and compliance.
  - Developed multiple APIs using **Django Rest Framework**, and FastAPI and implemented **dynamic invoice templates using Jinja**, allowing merchants level branding and PDF invoice generation.
  - Implemented **RAG Pipeline for Semantic invoice search** using **vector embeddings stored in Pinecone**, enabling finance teams to search invoices with natural language queries e.g. "vendor payments for January travel"
  - Introduced **feature flags** for controlled rollouts, A/B validation, and instant rollback of production features.
  - Implemented **canary-style deployments**, gradually exposing new features to subsets of users and validating releases using live metrics
  - Built **async background task processing pipelines** using **RabbitMQ, Celery, AWS SQS, AWS SNS, and AWS Lambda** for invoice processing, notifications, reconciliation and report generation tasks.
  - **Led and executed database migration, query optimization, and AWS RDS Aurora cluster scaling** (read replicas, connection pooling, slow query tuning), and improving performance at scale.
  - Deployed & managed CI/CD pipeline using **Github Actions**. Implemented RBAC & JWT and automated unit & integration testing using **Pytest** and **FactoryBoy**.
  - Deployed **ELK stack (Elastic Search, Logstash and Kibana)** for infra monitoring and analytics and Grafana and Prometheus for application performance metrics and alerting.
  - Deployed infrastructure using **Terraform (IaaC)**.
  - Leveraged **LLMs and Prompt Engineering** for development assistance, test case generation, and debugging, reducing manual overhead by ~40%. Used Cursor AI, WindSurf and Github Copilot
  - Led engineering team, managed sprint delivery, did code review, and mentored engineers

- **Assistant Manager (Data Scientist)**
  Ceipal Software Pvt. Ltd., Hyderabad — May. 2019 - Jun. 2022
  1. **AI Resume Ranking System**

- Built AI-powered resume ranking system using semantic similarity, NER, and embeddings (Skill2Vec, JD Classifier), reducing candidate screening time by 55% and improved ranking accuracy by 35%
- Implemented JD Parser, to extract similar job title and skills. Trained and fine-tuned model Skill2Vec, Job Description classifier, and Organization Industry extractor models using Word embedding and NER – Named Entity Recognition
- Developed and maintained a **skill taxonomy using DBpedia and SPARQL**, mapping extracted skills to hierarchical and related categories, enhancing semantic understanding of candidate skills and job descriptions.
- Model deployment using Python/FastAPI + MLflow/MLServer on Google Cloud enabling scalable, low latency inference.
- Designed and optimized Apache Solr schema, custom analyzers, synonym filters, and ranking configuration, collaborating with recruiters to fine-tune skill relevance, recency, and job title boosting – improving candidate search speed and accuracy by 35%.
- Tuned Solr scoring with field boosting (skills, job title, recency) using edismax parser to deliver recruiter preferred candidate ranking. Engineered near real-time indexing pipeline, optimized caching and query performance, and scaled Solr to deliver sub-second search results at production scale
- Tech Stack: Django, Keras, Pytorch, Docker, Huggingface, Numba, Solr, Neo4j/SPARQL, Pandas, BERT, Numpy, Prodigy, Redis, RabbitMQ, Software Development, NLTK, Spacy

2. Psychometric Data extraction from candidate interview video

- Developed **real-time video analytics models** (age, gender, emotion detection). Tech Used: Python, Fast API, Docker, PostgreSQL, Keras and OpenCV for psychometric profiling of candidates.

- **Associate Data Scientist**

Edelweiss Business Services Limited., Mumbai — Nov. 2017 - Mar. 2019

1. Cognitive Email Tagger for HROPS

- Built **Cognitive Email Tagger**, auto-classifying 5000+ employee's support queries/month into different categories - Payroll, Compliance, LMS etc., reduced manual efforts by 83%, and reduced response time by auto ticket generation & assignment to HROPS team.
- Prepared training data, did model training and validation, and deployed model using Flask
- Used Python, NLTK, Keras, Tensorflow, Scikit-Learn, Redis, Docker, Celery, PostgreSQL and RabbitMQ.

2. Company News Search Engine

- Developed **Company News Search Engine** Apache Solr (schema design, analyzers, synonym filters, boosting) + Scrapy, Python/Flask, NLTK, enabling real-time company-specific news retrieval and improving research efficiency for analysts using event driven architecture leveraging RabbitMQ, and Celery for Asynchronous Message Queue.
- Optimized Solr indexing and query performance, configured faceting, and tuned ranking parameters for better relevance using BM21 algorithm and eDisMax query parser.

3. Airline Invoice Automation

- Implemented Airline Invoice Ingestion engine handling 100+ invoice templates from different Airlines using Python, Pandas, pdfbox. Reduced invoice processing time from days to hours by 85%.

- **AI/NLP Research Intern**

Artivatic Data Labs Pvt Ltd., Bangalore — Jan. 2017 - Jun. 2017

1. Resume Summarizer

- Created **resume parser & summarizer**. Built resume parser, ingesting resumes in different formats – implemented job title extractor, and technical skills using Named Entity Recognition (NER), Word Sense Disambiguation (WSD). Tech Used: StanfordNLP, Apache Cassandra and REST API using Flask.

2. Credit Card Statement Parser

- Developed **credit card statement parser** with transaction categorization & visualization into different categories e.g. Entertainment, Travel etc. and visualization of categories. Tech: Python

EDUCATION

- **Pondicherry University, Puducherry** — Jul. 2014 - Jun. 2017
  Master of Computer Application (M.C.A) — CGPA: 8.3/10
- **Patna College, Patna** — Jul. 2010 - Jun. 2014
  Bachelor of Arts in Computer Application (B.C.A) — Percentage: 73.3%
- **Jawahar Navodaya Vidyalaya, Supaul** — May. 2010
  Senior Secondary Examination (CBSE) — Percentage: 61.4%

SKILL SETS

Languages: Python, TypeScript

Backend & Platform: Django, Flask, FastAPI, Async Processing, Event Driven System, Celery, RabbitMQ, AWS Lambda, AWS SQS
Database and Search: MySQL, PostgreSQL, Redis, Apache Cassandra, Solr, Neo4j, DynamoDB
Vector Database: Pincone, Qdrant DB
ML/AI: PyTorch, TensorFlow, HuggingFace, Spacy, Keras, NLTK, StanfordNLP, Apache Spark, Prodigy NLP, Pandas, MLFlow, MLServer, LangChain, LangGraph, CrewAI, LlamaIndex, and Deep Learning
Cloud & Infra: AWS, Docker/Kubernetes, Observability (ELK Stack, Grafana, Prometheus), CI/CD (Github Actions, CircleCI)
Open Source: Contribution to open source project **Scikit-Learn** and **Requests**.

STRENGTH

- Scaling products from zero-to-one
- Entrepreneurial mindset
- Quick-learner